Statistical Learning

Home Assignment 1
Question 1

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Ten-Year Trends in Grade 9 Math Pass Rates
and an Analysis of Economic, Political, and Demographic Factors Affecting These Rates in Dalarna County (2015–2023)

1. Introduction

Over the past two decades, Sweden's students' academic performance in main subjects such as mathematics has been closely monitored through international assessments. According to the Programme for International Student Assessment (PISA), Swedish 9-th graders have experienced a decline in mathematics scores, decreasing from just over 510 points in the early 2000s to a low of around 478 points in 2012, before going back to almost 500 points by 2018. Despite these national-level gains, some regions continue to lag behind the OECD average, raising concerns about the effectiveness of the efforts and changes related to improving education performance.

In this assignment, we treat the grade-9 math passing rate as a response variable and use regression for inference, understanding and quantifying how specific factors related to students impact the math result, across the 15 municipalities of Dalarna County over the period 2015–2023. Our main goals are:

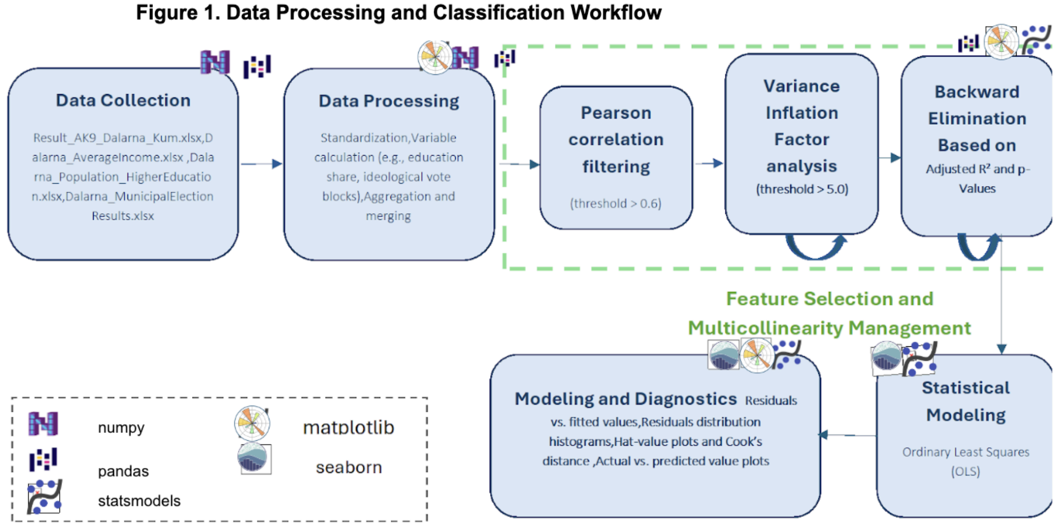
1. **Trend analysis:** Determine whether there is an overall upward or downward trend in passing rates within each municipality over this nine year period.
2. **Demographic, economic associations and Political effect::** Examine how local characteristics, such as average income, income inequality, and the share of people with high education, relate to math outcomes as well as assessing if parliament election results are linked to changes in student performance.

2. Methods and Data

2.1. Data Description

This study uses municipality-level data from Dalarna County, Sweden,

Figure 1. Data Processing and Classification Workflow



covering the years 2015 to 2024 which were obtained from open governmental or statistical sources. Four official datasets were used are:

-Educational outcomes: Grade 9 mathematics passing rates by municipality (Result_AK9_Dalarna_Kum.xlsx).

-**Socioeconomic indicators:** Average income, median income, and income inequality (Dalarna_AverageIncome.xlsx).

-Educational attainment: Share of population with higher education (Dalarna_Population_HigherEducation.xlsx).

-Political orientation: Municipal election results used to calculate ideological vote shares (Dalarna_MunicipalElectionResults.xlsx).

2.2. Data Cleaning

Preprocessing was conducted in Python using structured data classes. The data cleaning and key transformations included:

- · Calculation of higher education share as a percentage of the adult population.
- Municipality names and year formats were standardized across datasets. Missing values were addressed through forward-filling and removal of incomplete rows.
- · Aggregation of party-level election results into left- and right-leaning vote shares.

- · Expansion of election results to cover non-election years by forward-filling the previous election cycle's data.
- · Good to note that although the question asked for an analysis of the ten-year period (2015–2024), there were inconsistencies in the data, as some datasets covered different years. So, to make the statistical analysis more accurate and clear, we decided to use only the overlapping years common to all datasets, which are from 2015 to 2023.

The final dataset contained the following variables: Math_grade_rate_result as a dependent variable, Average_income, Median_income, Income_inequality, Higher_education_percentage, Education_share, Left_share, Right_share as well as population metrics.

2.3. Statistical Modeling

Considering the two main questions mentioned in the introduction section, we first began by identifying the type of problem.

For the first question, which relates to trend analysis, we are dealing with identifying patterns or describing behavior over time. Therefore, it is categorized as an inference problem, and we are supposed to answer it using linear regression.

The second question also involves describing the relationship between features (economic, demographic, and political factors) and the response variable. This is also an inference problem. We began by exploring various regression models to find the one that could provide a clear conclusion from the dataset.

We have chosen ordinary least squares (OLS) regression with municipality fixed effects. This approach allows us to compare each municipality to itself over time. Also instead of a separate dummy for every year, we just put Year in as one continuous number (2015, 2016, …, 2023) to catch the general trend without making the model too complex.

We chose OLS regression because it is:

1. **Easy to understand:** Each coefficient tells us exactly how much the math pass rate changes when a predictor moves by one unit.
2. **Great for answering inference questions:** OLS is the standard way to estimate and test relationships between variables, which fits our goal of understanding (not just

predicting) the effects of income, education, and politics.

3. **Flexible with fixed effects:** By adding a dummy (0/1) variable for each municipality, OLS lets us control for anything that stays the same in a town over time.
4. **Simple to check:** We can run easy diagnostics (like VIF for multicollinearity and residual tests for normality) to make sure that our findings can be trusted.

Model Implementation

The final model was trained using OLS regression via the statsmodels package. Diagnostic checks were performed to assess model fit, residual distributions, and leverage points. Plots included:

· Residuals vs. fitted values
· Residuals distribution histograms
· Hat-value plots and Cook’s distance
· Actual vs. predicted value plots

All modeling code and diagnostics were reproducible and executed via Python scripts.

Feature Selection and Multicollinearity Management

A feature selection process was implemented to enhance model validity and interpretability by systematically addressing multicollinearity and statistical insignificance. This process involved two main stages: (1) filtering for multicollinearity, and (2) performing backward elimination guided by statistical inference and model performance.

Multicollinearity Filtering

To ensure that the regression model was not distorted by interdependent predictors, multicollinearity was assessed and mitigated through a combined approach involving both correlation analysis and Variance Inflation Factor (VIF) diagnostics:

- · Pairwise Correlation Analysis: A Pearson correlation matrix was computed for all continuous independent variables. Pairs of variables exhibiting high correlation (|r| > 0.6) were flagged as redundant. From each pair, one variable was removed based on domain relevance or overall redundancy in the dataset.

Figure 2. Backward Elimination

```
Dropping Left_share with p-value = 0.9896 because Adj R²: 0.4447 → 0.4495
Dropping Municipality_Säter with p-value = 0.9239 because Adj R²: 0.4495 → 0.4542
Dropping Municipality_Orsa with p-value = 0.7667 because Adj R²: 0.4542 → 0.4585
Dropping Municipality_Ludvika with p-value = 0.7111 because Adj R²: 0.4585 → 0.4624
Dropping Population_growth_rate with p-value = 0.6717 because Adj R²: 0.4624 → 0.4661
Dropping Municipality_Vansbro with p-value = 0.5370 because Adj R²: 0.4661 → 0.4689
Keeping Municipality_Hedemora with p-value = 0.1176 because Adj R²: 0.4689
```

- · Variance Inflation Factor (VIF) Thresholding: Following correlation filtering, VIF scores were calculated to quantify the degree of linear dependency among the remaining predictors. A recursive procedure was applied: the variable with the highest VIF (above a threshold of 5.0) was removed, and the VIFs were recalculated iteratively. This continued until all remaining predictors had VIFs within acceptable bounds, ensuring that the retained variables contributed unique information to the model.

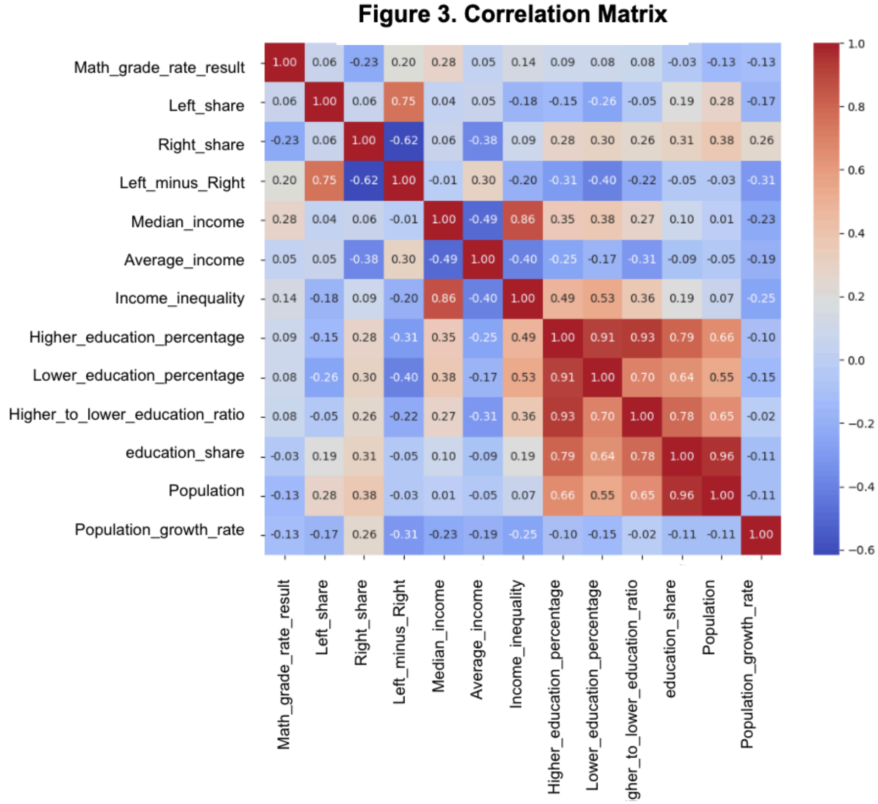
This step resulted in a refined set of predictors with minimized multicollinearity, suitable for reliable coefficient estimation and interpretation.

Backward Elimination Based on Adjusted R^2 and p-Values

Once a multicollinearity-free predictor set was established, backward elimination was conducted to remove statistically weak variables while preserving model quality. This was implemented using a custom iterative procedure:

- · An initial OLS regression model was fitted using all remaining predictors.

Figure 3. Correlation Matrix



- · In each iteration, predictors were evaluated based on their p-values, with the highest (least significant) variable identified for possible removal.
- · The model was re-estimated excluding that variable, and the resulting Adjusted R^2 was compared to the previous model.
- If Adjusted R^2 improved or remained unchanged, the variable was dropped.
- If Adjusted R^2 decreased, the variable was retained despite its p-value.

The process continued until all remaining variables had p-values below a threshold of 0.10 or were justified by their contribution to model performance.

This dual-phase feature selection—correlation/VIF pruning followed by statistically guided backward elimination—ensured a parsimonious yet informative model that balances simplicity, robustness, and explanatory power.

3. Results and Discussion

Results

An OLS regression was estimated for the grade-9 mathematics passing rate across 15 municipalities), using the following predictors: year, right-wing share, median household income, higher-education share, and municipality fixed-effects. The following table summarizes the result:

Figure 4. OLS Regression Results

```
==============================================================================
Dep. Variable:     Math_grade_rate_result   R-squared:                       0.524
Model:                                OLS   Adj. R-squared:                  0.469
Method:                     Least Squares   F-statistic:                     9.450
Date:                    Mon, 12 May 2025   Prob (F-statistic):           8.95e-14
Time:                            03:08:15   Log-Likelihood:                -372.50
No. Observations:                     135   AIC:                             775.0
Df Residuals:                         120   BIC:                             818.6
Df Model:                              14
Covariance Type:                nonrobust
=====================================================================================================
                                   coef    std err          t      P>|t|      [0.025      0.975]
-----------------------------------------------------------------------------------------------------
const                         1295.0870    393.927      3.288      0.001     515.138    2075.036
Year                            -0.6123      0.196     -3.126      0.002      -1.000      -0.224
Right_share                     -0.2407      0.076     -3.174      0.002      -0.391      -0.091
Median_income                    0.0302      0.048      0.626      0.533      -0.065       0.126
Higher_education_percentage      2.4392      0.401      6.079      0.000       1.645       3.234
Municipality_Borlänge           -3.9408      1.696     -2.324      0.022      -7.298      -0.584
Municipality_Falun             -14.8586      3.500     -4.245      0.000     -21.789      -7.928
Municipality_Gagnef              2.8294      1.635      1.731      0.086      -0.407       6.066
Municipality_Hedemora            2.3636      1.499      1.576      0.118      -0.605       5.332
Municipality_Leksand            -9.1148      2.285     -3.989      0.000     -13.639      -4.591
Municipality_Malung-Sälen        8.5644      1.800      4.758      0.000       5.001      12.128
Municipality_Mora               -9.4138      1.945     -4.840      0.000     -13.265      -5.563
Municipality_Rättvik             5.0324      1.551      3.245      0.002       1.962       8.103
Municipality_Smedjebacken        7.2083      1.739      4.146      0.000       3.766      10.650
Municipality_Älvdalen            8.7170      1.762      4.949      0.000       5.229      12.205
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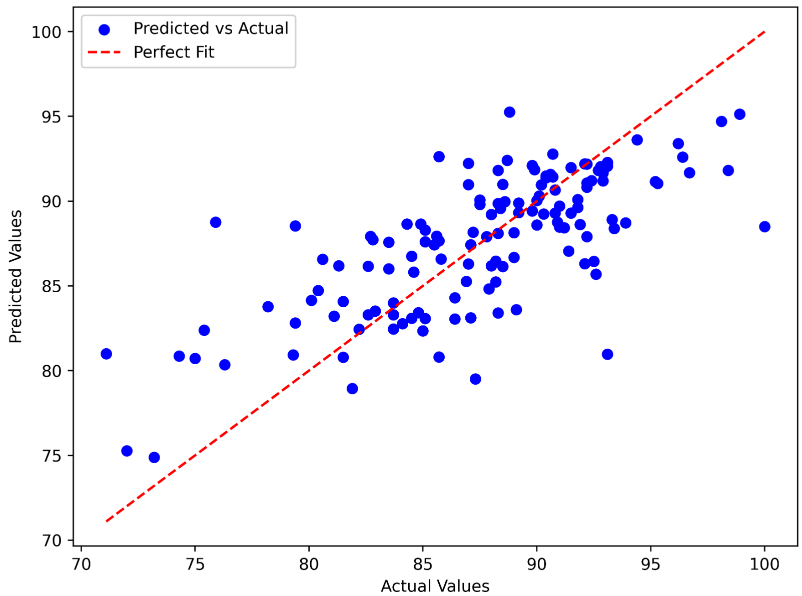
Figure 5. Actual vs. Predicted Grade-9 Math Pass Rates



Discussion

Overall Trend

The negative Year coefficient (–0.61, $p = 0.002$) shows pass rates have fallen by about 0.6 points each year, even though national scores have improved. This points to local issues slowing progress.

Education Level

The highest impact comes from adult education: a 1-point rise in the share of residents with a university degree leads to a 2.4-point increase in pass rates ($p < 0.001$).

Political Influence

Each 1-point increase in right-wing vote share corresponds to a 0.24-point drop in pass rates ($p = 0.002$). While it's hard to conclude, it can be interpreted that local political priorities may influence school resources.

Income

Median income has no clear effect once we account for education level and politics ($p = 0.533$). This means that education still matters the most.

Local Differences

Even after these factors, big gaps remain between towns: Falun (–14.9 points), Leksand (–9.1), and Mora

(–9.4) score below the baseline, while Malung–Sälen (+8.6), Älvdalen (+8.7), Smedjebacken (+7.2), and Rättvik (+5.0) score above. These differences likely come from unmeasured local strengths or challenges that need further investigation.

Insights about future Strategies

- **Invest in Education:** Programs to raise education levels among residents are likely the most powerful lever for improving student math outcomes.
- **Engage Political Parties:** Prepare proposals for political powers to help align resources and policies with educational goals.
- **Act on Yearly Declines:** Perform a review of teaching methods and other relevant factors to change the downward trend.
- **Start with underperformers:** Finally, clear gaps between towns mean we need local support efforts, learn from the top-performing municipalities and focus extra help on those still struggling.

Together, these steps are the best way to lift math pass rate across Dalarna.

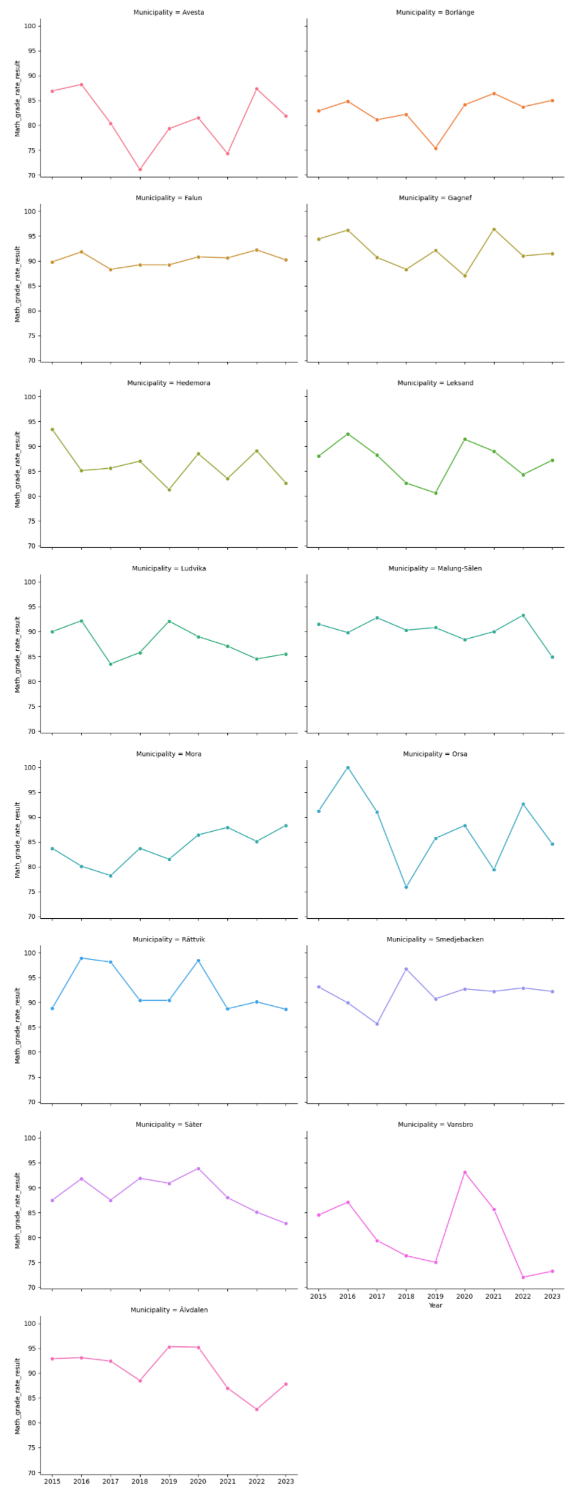


Figure 6. Grade-9 Math Pass Rates by Municipality